relation $Q = \frac{x^2 y^3}{z^1}$. The percentage errors in x, y and z are 2%, 3% and 1% respectively. Find the percentage error in Q.

Solution:

$$\text{Let, } Q = \frac{x^2 y^3}{z}.$$

$$\text{It is given, } \frac{\Delta x}{x} = 2\% \quad \frac{\Delta y}{y} = 3\% \quad \frac{\Delta z}{z} = 1\%$$

$$\frac{\Delta Q}{Q} = 2\left(\frac{\Delta x}{x}\right) + 3\left(\frac{\Delta y}{y}\right) + 1\left(\frac{\Delta z}{z}\right)$$

$$= 2(2\%) + 3\,(3\%) + 1(1\%)$$

$$\frac{\Delta Q}{Q} = 4\% + 9\% + 1\% = 14\%$$

5. The mass and volume of a body are found to be 4±.03 kg and 5±.01 m^3 respectively. Then find the maximum possible percentage error in density.

Solution:

$$\text{Mass m} = 4 \pm 0.03 \text{ kg (m} + \Delta\text{m)}$$

$$\text{Volume V} = 5 \pm .01 \text{ m}^3 \text{ (V} + \Delta\text{V)}$$

$$\text{Density} = ?$$

$$\text{Error in mass} = \frac{\Delta m}{m} = \frac{0.03}{4} \times 100$$

$$= 0.75\%$$

$$\text{Error in volume} = \frac{\Delta V}{V} = \frac{0.01}{5} \times 100$$

$$= 0.2\%$$

$$\text{Density} = \frac{mass}{volume}.$$

Error in density = error in mass + error in volume

$$= 0.75\% + 0.2\% = 0.95\%$$

6. Using a Vernier Callipers, the length of a cylinder in different measurements is found to be 2.36 cm, 2.27 cm, 2.26 cm, 2.28 cm, 2.31 cm, 2.28 cm and 2.29 cm. Find the mean value, absolute error, the relative error and the percentage error of the cylinder.

Solution:

The given readings are 2.36 cm, 2.27 cm, 2.26 cm, 2.28 cm, 2.31 cm, 2.28 cm and 2.29 cm

$$\text{The Mean value } \bar{l} = \frac{2.36 + 2.27 + 2.26 + 2.28 + 2.31 + 2.28 + 2.29}{7}$$

$$\frac{16.05}{7} = 2.29 \text{ cm}$$

Absolute errors in the measurement are,

$$|\Delta l_1| = |2.29 - 2.36| = 0.07$$
$$|\Delta l_2| = |2.29 - 2.27| = 0.02$$
$$|\Delta l_3| = |2.29 - 2.26| = 0.03$$
$$|\Delta l_4| = |2.29 - 2.28| = 0.01$$
$$|\Delta l_5| = |2.29 - 2.31| = 0.02$$
$$|\Delta l_6| = |2.29 - 2.28| = 0.01$$
$$|\Delta l_7| = |2.29 - 2.29| = 0.00$$

Mean Absolute error

$$\Delta l_{mean} = \frac{0.07 + 0.02 + 0.03 + 0.01 + 0.02 + 0.01 + 0.00}{7}$$

$$= \frac{.16}{7} = .02$$

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